

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · COMBINED SCIENCE

COMBINED SCIENCE 5006/2

PAPER 2 November 2024

2 hours

PAPER 2 — STRUCTURED QUESTIONS — ANSWER ALL — 2 HOURS

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2024 5006/2.

Additional materials: Answer all questions. The number of marks is given in brackets [].

INSTRUCTIONS: Write your answers in the spaces provided. You may use a calculator.

1. (a) State one adaptation of a palisade mesophyll cell and link it to photosynthesis. [2]<<
>>(b) State one adaptation of a red blood cell for oxygen transport. [2]<<
>>(c) Describe how a sperm cell is adapted to reach and fertilise an egg. [3]<<
>>(d) Explain why ciliated cells line the trachea. [3] **[10]**

2. (a) Write the balanced symbol equation for photosynthesis. [3]<<
>>(b) Name the green pigment and the organelle where it is found. [2]<<
>>(c) State two factors that can limit the rate of photosynthesis. [2]<<
>>(d) Describe a test for starch in a destarched leaf that has been exposed to light. [3] **[10]**

3. (a) Name the four chambers of the human heart. [2]<<
>>(b) Name the blood vessel that carries oxygenated blood from the lungs to the heart. [1]<<
>>(c) Describe the path of oxygenated blood from the lungs to a capillary in the body. [4]<<
>>(d) Give two differences between an artery and a vein. [3] **[10]**

4. (a) Name the type of pathogen that causes malaria. [1]<<
>>(b) Name the vector of malaria. [1]<<
>>(c) Describe how malaria is transmitted from an infected person to a healthy person. [4]<<
>>(d) Suggest two methods of controlling the spread of malaria. [4] **[10]**

5. (a) Describe how an ionic bond forms between sodium and chlorine. [3] <<
>>(b) Describe a covalent bond. [2] <<
>>(c) Explain why sodium chloride has a high melting point. [3] <<
>>(d) Explain why chlorine gas, Cl_2 , has a low melting point. [2]
[10]
6. (a) Write a word equation for hydrochloric acid reacting with calcium carbonate. [3] <<
>>(b) Describe the test for carbon dioxide. [2] <<
>>(c) Universal indicator is added to the acid before the carbonate. Describe the colour change as the acid is used up. [2] <<
>>(d) Explain why farmers add lime (calcium carbonate) to some soils. [3] **[10]**
7. (a) State the principle of conservation of energy. [2] <<
>>(b) A 4 kg mass is lifted through 10 m. Take $g = 10 \text{ N/kg}$. Calculate the gain in gravitational potential energy. [3] <<
>>(c) Name the energy transfer that occurs in a battery-powered torch when it is switched on. [2] <<
>>(d) Explain how heat is transferred through the metal wall of a kettle by conduction. [3] **[10]**
8. (a) State the law of reflection. [2] <<
>>(b) A ray in air hits a glass block. Describe what happens to its speed and direction. [3] <<
>>(c) Explain why sound cannot travel through a vacuum, but light can. [3] <<
>>(d) State how pitch and loudness of a sound relate to the wave. [2] **[10]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).